

AVONDALE 2 N, PA, USA

WMO: 725160

Lat: 39.8593N

Lon: 75.7861W

Elev: 400

StdP: 14.48

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 03761

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.4	13.5	-5.7	4.1	10.0	-1.2	5.2	14.2	17.0	28.6	15.9	28.2	5.1	N/A	0.381

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.2	88.9	74.9	86.5	73.4	84.3	72.1	77.1	85.3	75.6	83.4	74.4	81.3	4.2	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.6	131.5	81.6	73.3	125.4	79.5	72.2	120.8	78.2	40.7	85.2	39.3	83.6	38.1	81.4	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.7	12.7	10.7	DB	3.8	93.2	5.1	3.1	0.1	95.4	-2.9	97.2	-5.8	98.9	-9.6	101.1	(4)
			WB	1.8	79.8	4.8	1.1	-1.6	80.6	-4.5	81.3	-7.2	81.9	-10.7	82.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	52.9	30.4	32.4	40.7	51.7	61.8	69.5	74.1	71.8	65.9	55.1	43.7	36.5	(6)
		DBStd	17.13	10.15	10.06	9.65	8.68	7.61	5.56	4.66	4.71	6.60	8.33	8.07	8.97	(7)
		HDD50	2204	610	499	318	79	4	0	0	0	43	221	430		(8)
		HDD65	5240	1071	912	754	408	155	21	1	69	324	638	883		(9)
		CDD50	3273	4	7	29	130	371	586	747	676	478	200	33	12	(10)
		CDD65	835	0	0	0	9	57	157	283	216	97	16	0	0	(11)
		CDH74	6556	0	0	6	109	486	1259	2361	1584	660	91	0	0	(12)
		CDH80	1817	0	0	0	18	112	327	774	405	168	13	0	0	(13)
(14)	Wind	WSAvg	3.6	4.7	5.1	4.9	4.4	2.9	2.9	2.3	2.2	2.4	3.2	3.9	4.3	(14)
(15)	Precipitation	PrecAvg	46.9	3.7	2.6	4.4	3.8	3.5	4.5	4.1	4.0	4.8	4.0	3.4	3.9	(15)
(16)		PrecMax	64.3	6.1	5.8	7.6	7.4	7.0	11.3	7.8	7.7	12.8	10.4	5.0	8.1	(16)
(17)		PrecMin	34.7	1.6	0.5	0.5	1.7	1.4	1.9	1.1	1.6	0.4	0.7	1.0	0.9	(17)
(18)		PrecStd	8.2	1.3	1.1	1.7	1.5	1.5	2.6	1.8	1.5	3.2	2.5	1.2	1.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	61.8	67.5	73.3	83.0	87.0	91.0	93.0	90.1	88.7	80.9	69.2	65.2		(19)
(20)		MCWB	57.4	57.2	60.2	65.9	71.6	74.8	78.0	76.9	73.8	69.0	59.6	61.5		(20)
(21)		DB	56.2	58.9	66.7	77.1	83.3	86.8	89.3	86.8	85.0	76.4	65.1	59.9		(21)
(22)		MCWB	52.2	52.6	56.7	63.4	69.8	72.1	75.6	74.1	71.9	68.2	58.4	57.6		(22)
(23)		DB	50.4	53.0	61.0	72.2	79.5	83.8	86.9	84.3	81.1	72.4	60.9	55.2		(23)
(24)		MCWB	45.5	46.3	53.2	61.3	67.8	71.0	74.5	72.4	70.1	65.1	53.9	51.9		(24)
(25)		DB	45.5	47.0	56.2	67.8	75.9	81.0	84.4	82.1	77.6	68.7	57.3	50.2		(25)
(26)		MCWB	41.4	41.6	48.5	57.8	65.4	69.3	73.0	71.4	68.2	62.3	51.4	46.2		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	58.4	60.3	62.1	68.3	74.2	76.6	80.0	78.3	76.5	72.6	63.9	62.5		(27)
(28)		MCDB	59.6	66.2	70.5	75.3	84.0	86.5	89.7	86.0	84.4	76.6	66.9	64.4		(28)
(29)		WB	51.5	54.0	58.3	65.7	72.0	74.5	77.9	76.4	74.7	69.7	60.0	58.9		(29)
(30)		MCDB	54.1	57.2	64.7	73.1	80.3	83.2	86.6	83.3	81.2	74.5	63.7	60.4		(30)
(31)		WB	46.1	47.6	54.0	62.3	70.0	73.0	76.2	74.8	73.0	67.3	56.2	54.1		(31)
(32)		MCDB	49.2	52.9	60.5	70.6	77.6	80.3	84.4	81.1	78.3	71.8	59.9	56.2		(32)
(33)		WB	42.1	42.9	49.5	59.1	68.1	71.4	74.9	73.1	71.2	64.1	52.7	48.5		(33)
(34)		MCDB	45.3	47.9	55.7	65.6	74.4	77.6	82.3	78.4	75.5	68.3	56.5	50.7		(34)
(35)	Mean Daily Temperature Range	MDBR	14.9	15.6	17.5	19.8	19.1	18.7	18.2	18.0	18.6	18.3	17.1	14.4		(35)
(36)		5% DB	21.0	23.2	25.1	25.5	23.0	21.6	20.3	19.7	21.9	21.5	21.8	18.9		(36)
(37)		MCWBR	18.4	17.2	17.2	13.7	11.5	9.9	8.4	8.9	9.8	12.2	15.8	15.7		(37)
(38)		5% WB	20.1	22.5	23.4	22.1	20.6	18.1	17.7	16.5	16.7	16.8	18.8	16.9		(38)
(39)		MCWBR	18.1	18.1	17.1	14.2	11.1	9.5	8.5	8.3	8.9	11.6	16.1	15.9		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.309	0.318	0.356	0.397	0.439	0.503	0.508	0.506	0.418	0.358	0.323	0.306		(40)
(41)		taud	2.451	2.409	2.375	2.280	2.196	2.055	2.057	2.049	2.269	2.444	2.508	2.503		(41)
(42)		Ebn at Noon	272	286	285	279	268	250	247	244	260	267	266	265		(42)
(43)		Edh at Noon	25	30	35	41	46	53	52	51	39	29	24	23		(43)
(44)	All-Sky Solar Radiation	RadAvg	610	893	1179	1528	1708	1876	1897	1661	1367	984	699	529		(44)
(45)		RadStd	52	102	91	132	207	146	90	101	160	104	62	56		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (80 neighbors)	+0.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+229	+133

Nomenclature: See separate page